

Engineering Mechanics: ME101

Lecture 14: Statics



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SCREW JACK



- ✓ **Basic use:** Raise Heavy weights by small distances.
- ✓ Screws are used for fastening and for transmitting power or motion
- ✓ **Square-threaded** screws are frequently used in jacks, presses, and other mechanisms.
- ✓ Different Types of Threads which can be used are- Square, V-Shape, Buttress, ACME, Whitworth, etc.
- ✓ **For Screw Jacks**, Square threads are found to be most efficient.

Screw

❖ Square Threaded Screws

- ✓ Used for fastening and for transmitting power or motion
- ✓ Square threads are more efficient.
- ✓ Friction developed in the threads largely determines the action of the screw.

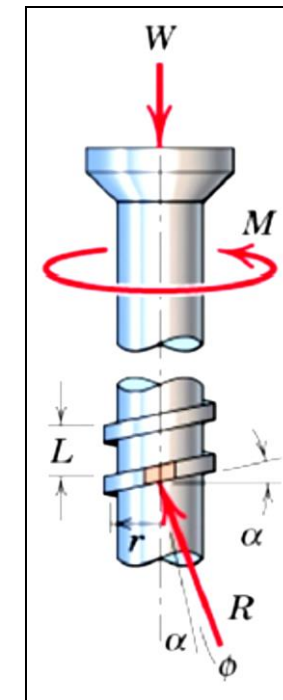
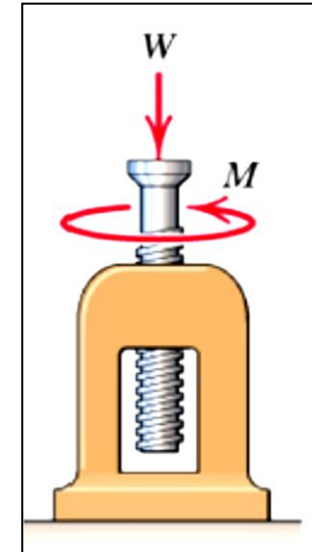
❖ FBD of the Screw: R exerted by the thread of the jack frame on a small portion of the screw thread is shown,

- ✓ Lead = L = advancement per revolution.
- ✓ L = Pitch (for single threaded screw).
- ✓ L = $2 \times$ Pitch (for double threaded screw where twice advancement per revolution).

❖ Pitch = axial distance between adjacent threads on a helix or screw,

Mean Radius = r ;

α = Helix Angle



Thread Terminologies

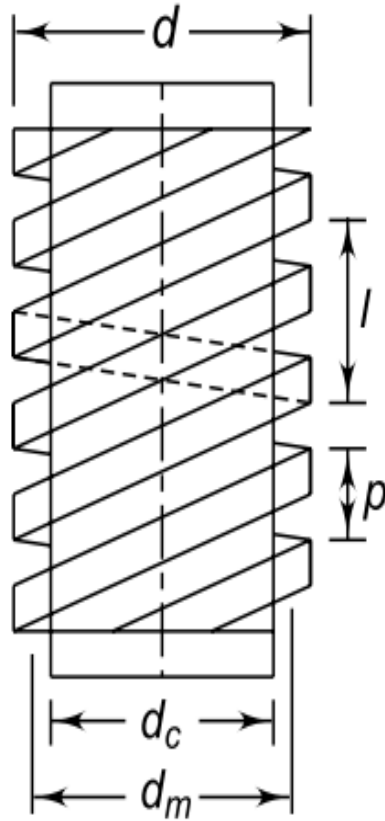
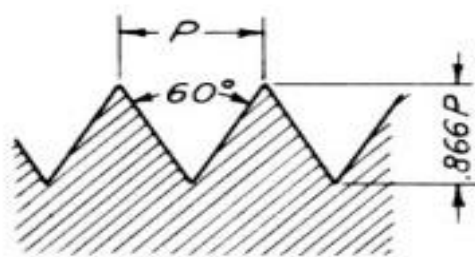


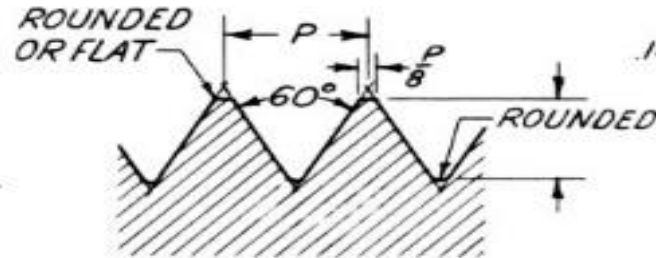
Fig. 6.4 Terminology of Power Screw

- **(i) Pitch** The pitch is defined as the distance measured parallel to the axis of the screw from a point on one thread to the corresponding point on the adjacent thread.
- **(ii) Nominal Diameter** Nominal diameter is the largest diameter of the screw. It is also called *major diameter*.
- **(iii) Core Diameter** The core diameter is the smallest diameter of the screw thread. It is also called *minor diameter*.

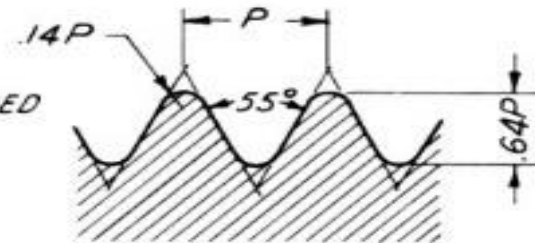
Different Types of Threads



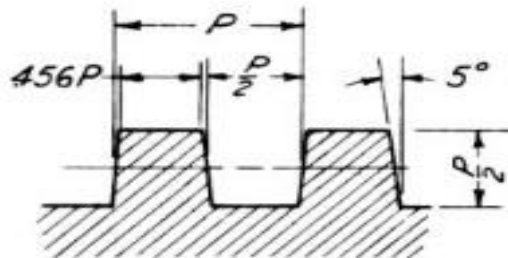
SHARP V



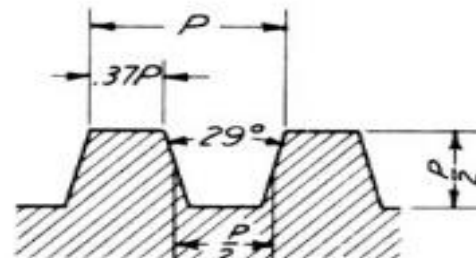
UNIFIED



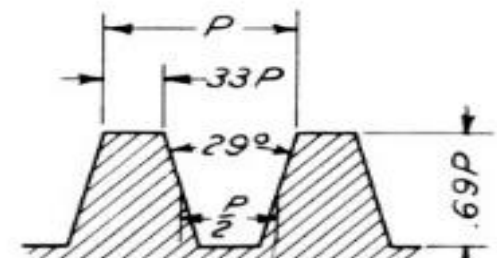
WHITWORTH



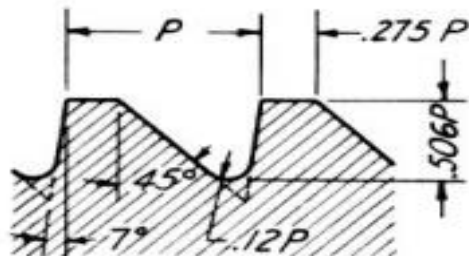
MODIFIED SQUARE
10° INCLUDED ANGLE



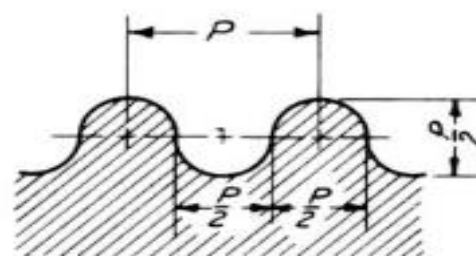
ACME



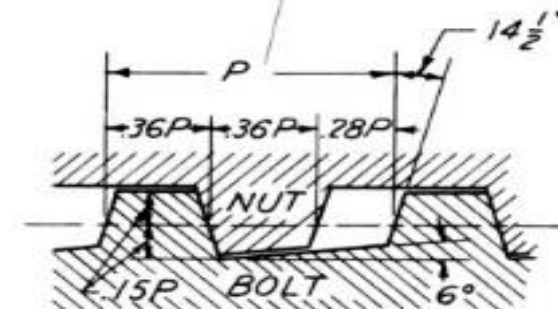
B & S WORM



MODIFIED BUTTRESS



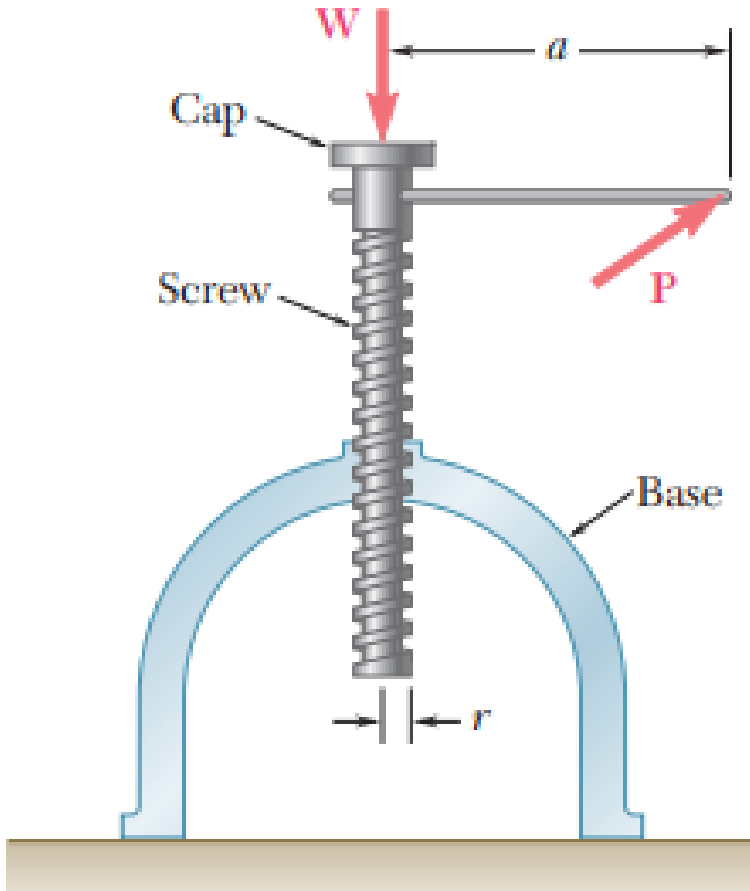
KNUCKLE



DARDELET

Screw threads.

Schematic of Screw Jack



- Screw Carries Load “ W ” which is supported by the Base
- Contact between screw and base takes place along a portion of their threads
- Applying a force P on the handle, the screw can be made to turn and to raise the load W .
- Lifting of Weight “ W ” in Screw Jack can be compared with a Weight “ W ” being lifted on an angular incline.
- Lead of a screw is the Lift of Weight because of one complete revolution of Screw.
- Lead – 1) Single Start thread = pitch
2) Multi start thread = $n \times \text{pitch}$

Different Moments & Loads acting on the Screw

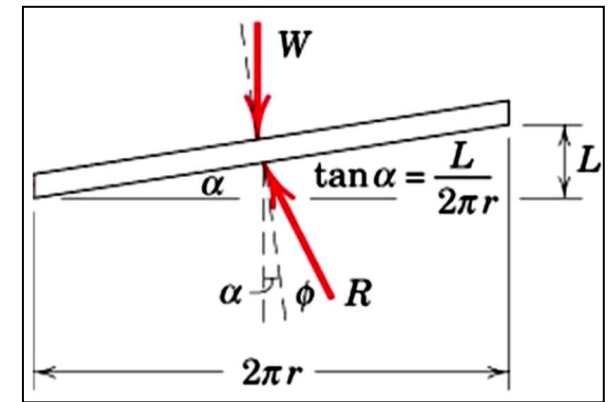
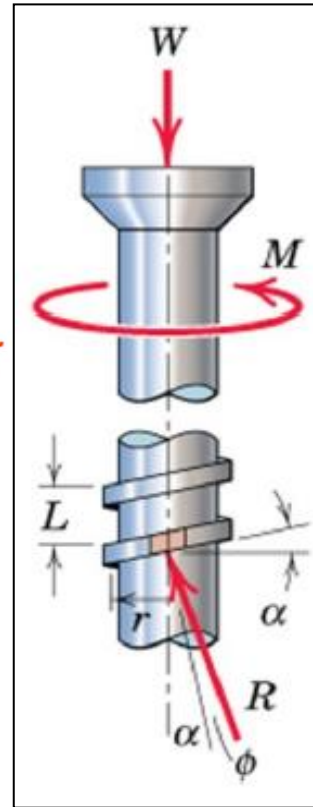
- ✓ Similar reactions exist on all segments of the screw threads. Analysis similar to **block on inclined plane** since friction force does not depend on area of contact.
- ✓ Thread of base can be “**unwrapped**” and shown as straight line. Slope is $2(\pi)r$ horizontally and lead L vertically.

If M is just sufficient to turn the screw $\rightarrow$ Motion Impending
Angle of friction = ϕ (made by R with the axis normal to the thread)
 $\rightarrow \tan \phi = \mu$

Moment of R @ vertical axis of screw = $R \sin(\alpha + \phi) r$
 $\rightarrow$ Total moment due to all reactions on the thread = $\sum R \sin(\alpha + \phi) r$
 $\rightarrow$ Moment Equilibrium Equation for the screw:
 $\rightarrow M = [r \sin(\alpha + \phi)] \sum R$

Equilibrium of forces in the axial direction: $W = \sum R \cos(\alpha + \phi)$
 $\rightarrow W = [\cos(\alpha + \phi)] \sum R$

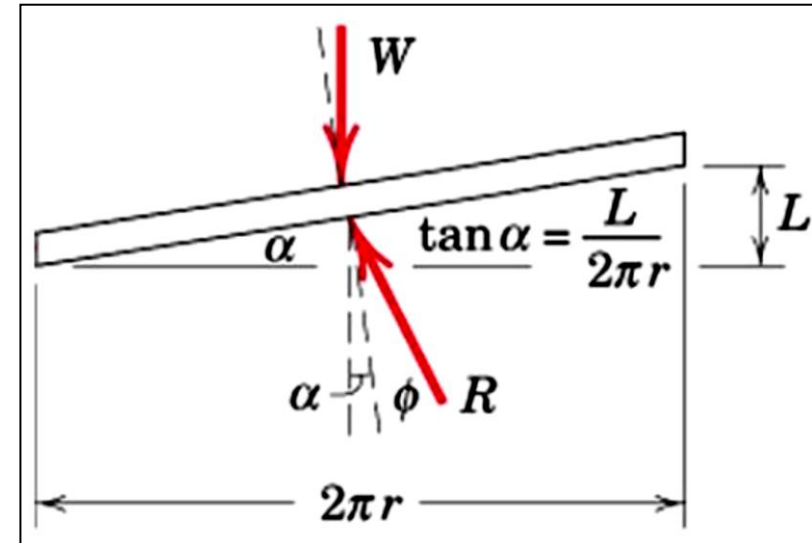
Finally $\rightarrow M = W r \tan(\alpha + \phi)$



Helix Angle of Screw

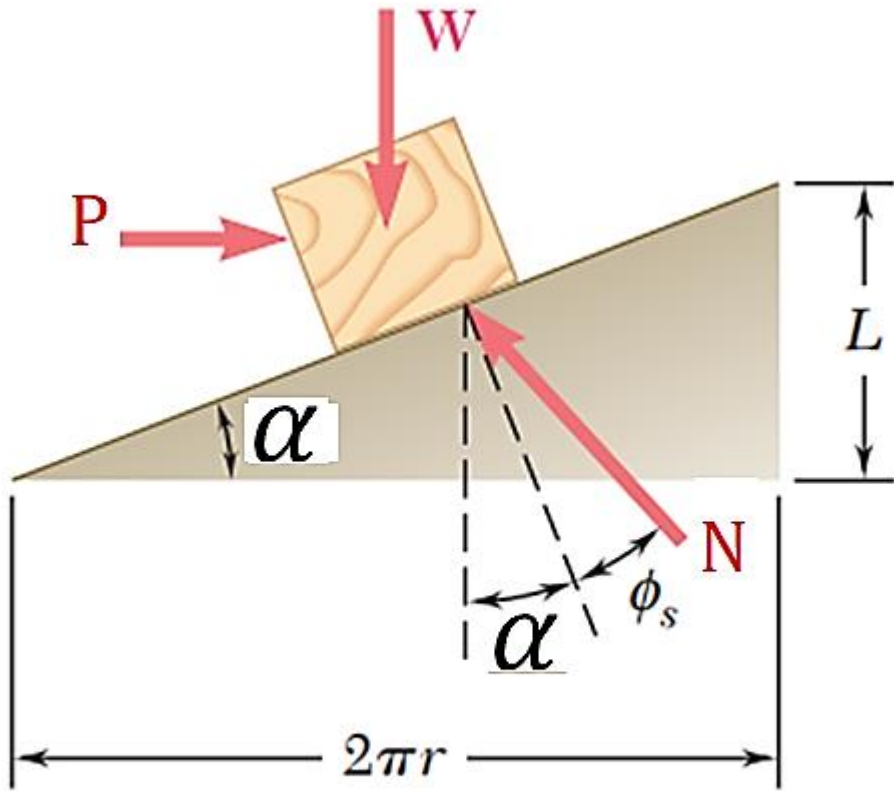
Helix angle α can be determined by unwrapping the thread of the screw for one complete turn

$$\alpha = \tan^{-1} (L/2\pi r)$$



Analysis of Screw Jack....(contd)

□ Unwrapping of thread involved in one rotation of screw

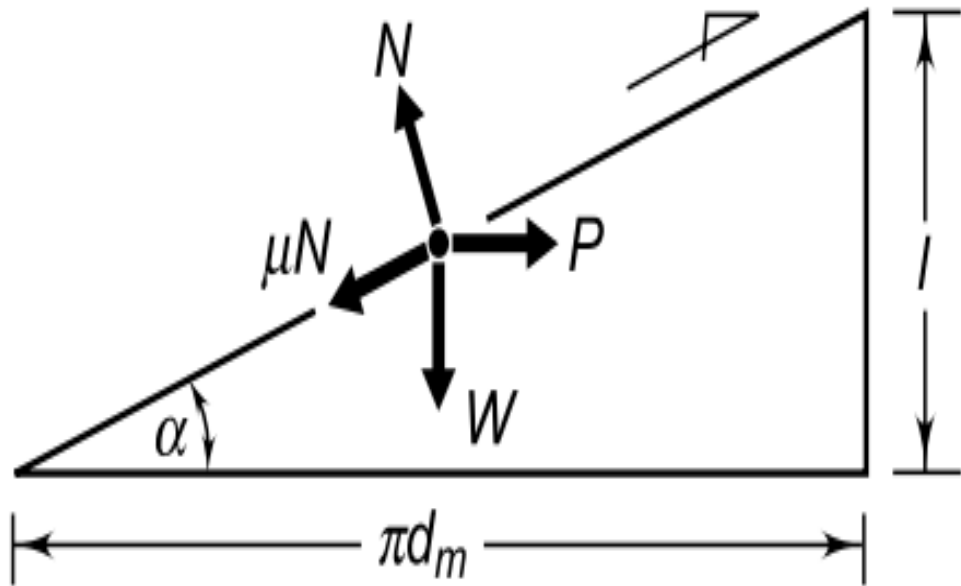


(a) Impending motion upward

- Right angle triangle is obtained as shown.
- Height of right angled triangle = LEAD of screw
- Base of Triangle = $2 \times \pi \times r$
 r = mean radius of thread

$$\tan \alpha = \frac{L}{2\pi r} \quad \text{Where "}\alpha\text{" is called as } \underline{\text{Lead Angle}}$$

Analysis for Lifting Load



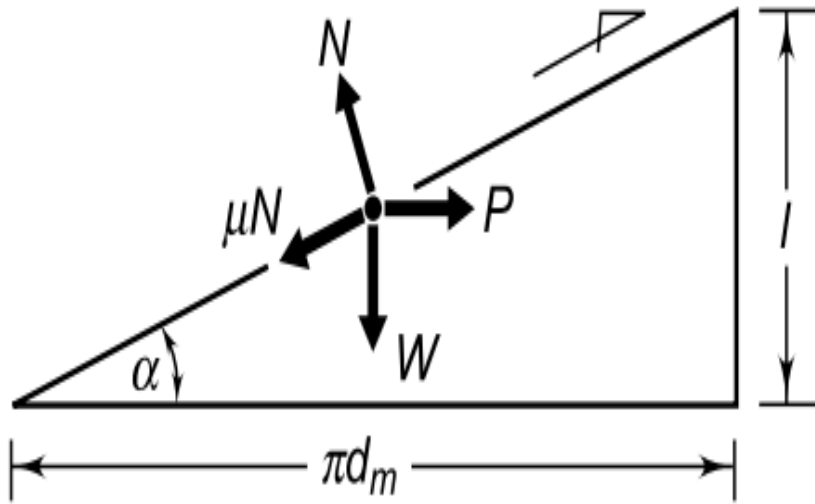
(i) **Load W :** It always acts in a vertically downward direction.

(ii) **Normal Reaction N :** It acts perpendicular (normal) to the inclined plane.

(iii) **Frictional Force μN :** Frictional force acts opposite to the motion. Since the load is moving up the inclined plane, frictional force acts along the inclined plane in the downward direction.

(iv) **Effort P :** Effort P acts perpendicular to the load W . It may act towards the right or towards the left. It should act towards the right to overcome the friction and raise the load.

Analysis for Lifting Load



- Horizontal Force Balance

$$P = \mu N \cos \alpha + N \sin \alpha$$
- Vertical force balance

$$W = N \cos \alpha - \mu N \sin \alpha$$
- Solving above equations,

$$P = \frac{W(\mu + \tan \alpha)}{(1 - \mu \tan \alpha)}$$

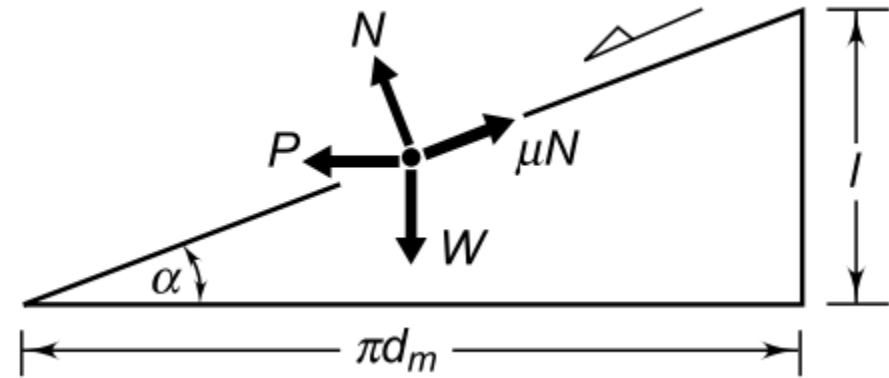
- Substituting, $\mu = \tan \phi$ in above equation, it can be simplified as given below :

$$P = W \tan(\phi + \alpha)$$

- Torque required to raise the load

$$M_t = \frac{P d_m}{2} = \frac{W d_m \tan(\phi + \alpha)}{2}$$

Analysis of Lowering Load



- Horizontal Force Balance

$$P = \mu N \cos \alpha - N \sin \alpha$$

- Vertical force balance

$$W = N \cos \alpha + \mu N \sin \alpha$$

- Solving above equations,

$$P = \frac{W(\mu - \tan \alpha)}{(1 + \mu \tan \alpha)}$$

- Substituting, $\mu = \tan \phi$ in above equation, it can be simplified as given below:

$$P = W \tan(\phi - \alpha)$$

- Torque required to lower the load

$$M_t = \frac{P d_m}{2} = \frac{W d_m \tan(\phi - \alpha)}{2}$$

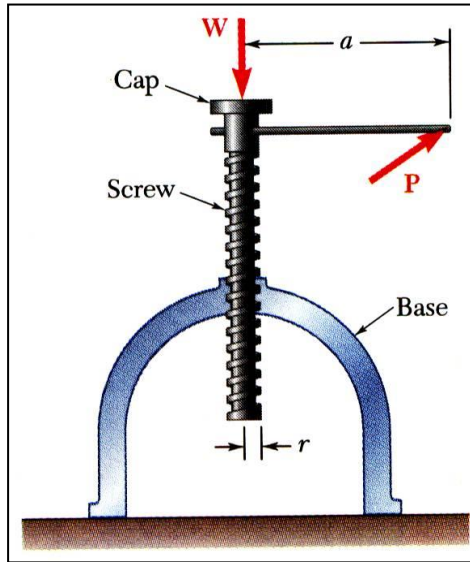
Self Locking Screw

- Torque required to lower the load

$$M_t = \frac{P d_m}{2} = \frac{W d_m \tan(\phi - \alpha)}{2}$$

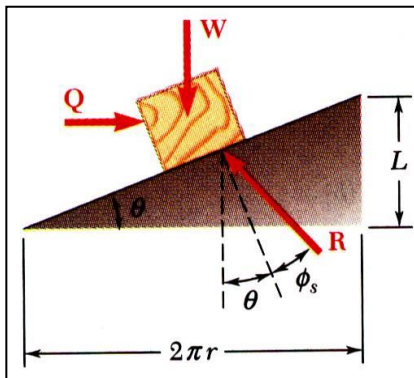
- If $\phi < \alpha$, value of torque required to lower load is negative. This implies that, zero effort will be required to lower the load and it will lower automatically. This is undesirable. It is called as “Overhauling”.
- If $\phi \geq \alpha$, a net positive torque is required to lower load, and it will not lower automatically. This is called **SELF LOCKING**.

Square-Threaded Screws

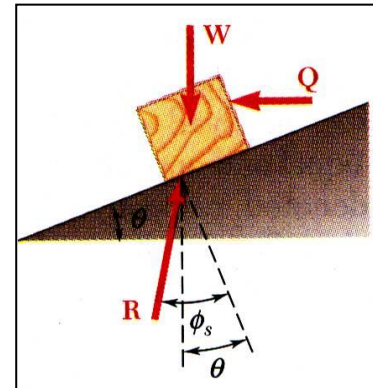


- **Square-threaded screws** frequently used in jacks, presses, etc. Analysis similar to block on inclined plane. Recall friction force does not depend on area of contact.
- Thread of base has been “unwrapped” and shown as straight line. Slope is $2(\pi)r$ horizontally and lead L vertically.
- Moment of force Q is equal to moment of force P .

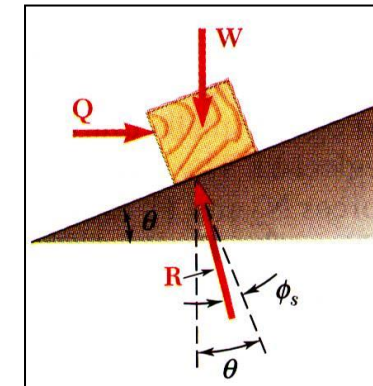
$$Q = Pa/r$$



- Impending motion upwards. Solve for Q .

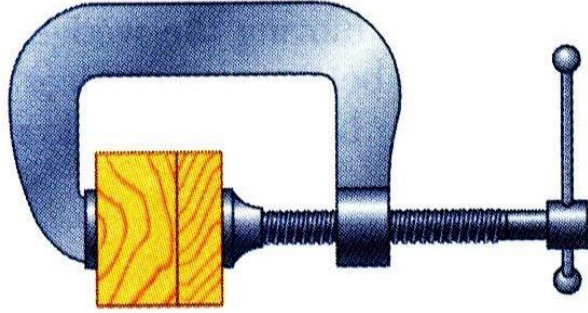


- $\phi_s > \theta$, Self-locking, solve for Q to lower load.



- $\phi_s > \theta$, Non-locking, solve for Q to hold load.

Sample Problem

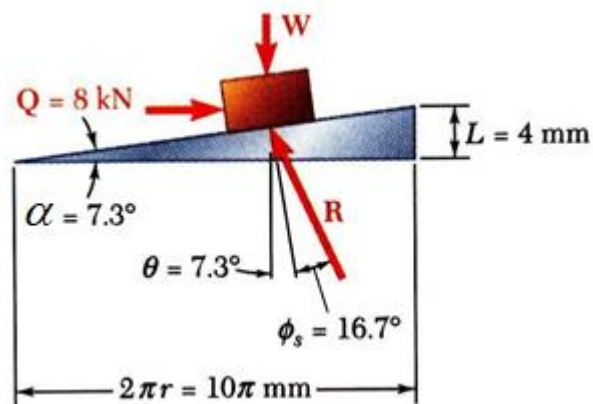


- A clamp is used to hold two pieces of wood together as shown. The clamp has a double square thread of mean diameter equal to 10 mm with a pitch of 2 mm. The coefficient of friction between threads is $\mu_s = 0.30$.
- ✓ If a maximum torque of 40 N*m is applied in tightening the clamp, determine (a) the force exerted on the pieces of wood, and (b) the torque required to loosen the clamp.

SOLUTION

- Calculate lead angle and pitch angle.
- Using block and plane analogy with impending motion up the plane, calculate the clamping force with a force triangle.
- With impending motion down the plane, calculate the force and torque required to loosen the clamp.

Sample Problem



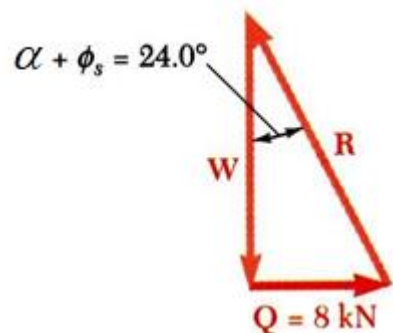
SOLUTION

- Calculate lead angle and pitch angle. For the double threaded screw, the lead L is equal to twice the pitch.

$$\tan \alpha = \frac{L}{2\pi r} = \frac{2(2 \text{ mm})}{10\pi \text{ mm}} = 0.1273 \quad \alpha = 7.3^\circ$$

$$\tan \phi_s = \mu_s = 0.30 \quad \phi_s = 16.7^\circ$$

- Using block and plane analogy with impending motion up the plane, calculate clamping force with force triangle.

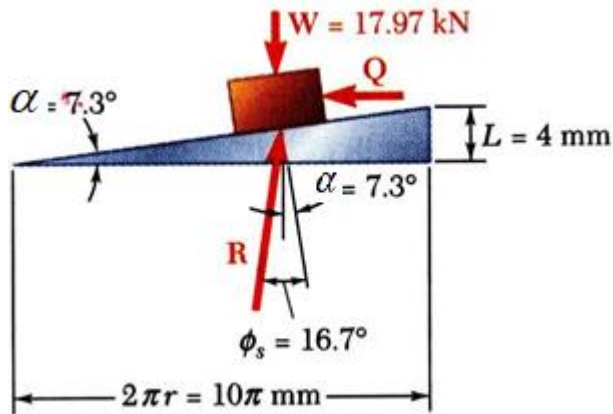


$$Qr = 40 \text{ N} \cdot \text{m} \quad Q = \frac{40 \text{ N} \cdot \text{m}}{5 \text{ mm}} = 8 \text{ kN}$$

$$\tan(\alpha + \phi_s) = \frac{Q}{W} \quad W = \frac{8 \text{ kN}}{\tan 24^\circ}$$

$$W = 17.97 \text{ kN}$$

Sample Problem



- With impending motion down the plane, calculate the force and torque required to loosen the clamp.

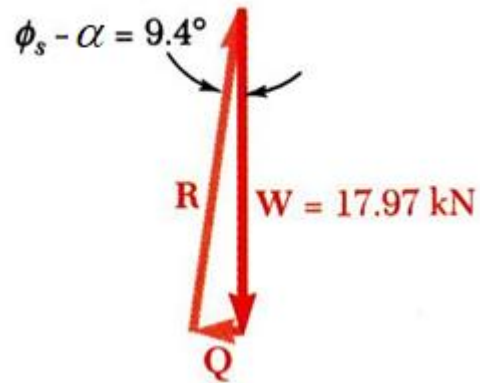
$$\tan(\phi_s - \alpha) = \frac{Q}{W} \quad Q = (17.97 \text{ kN}) \tan 9.4^\circ$$

$$Q = 2.975 \text{ kN}$$

$$\text{Torque} = Qr = (2.975 \text{ kN})(5 \text{ mm})$$

$$= (2.975 \times 10^3 \text{ N})(5 \times 10^{-3} \text{ m})$$

$$\text{Torque} = 14.87 \text{ N} \cdot \text{m}$$



Thrust Bearing & Disk friction

❖ Friction between circular surfaces under distributed normal pressure occurs in pivot bearings, clutch plates and disk brakes. To examine these applications,

- p is the normal pressure at any location
- frictional force acting on an elemental area = $\mu p dA$, where μ is the friction coefficient and dA is the area $r dr d\theta$

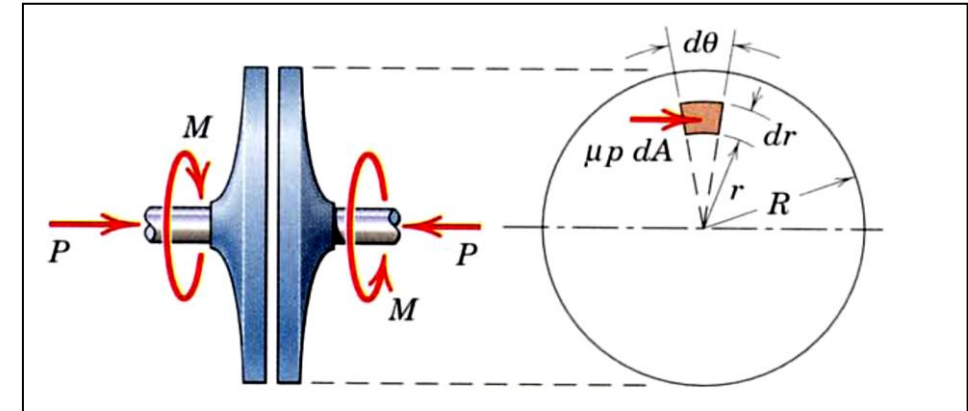
❖ Moment of the elemental friction,

$$M = \int \mu p r dA$$

❖ If p is uniform over the entire surface

$$\pi R^2 p = P.$$

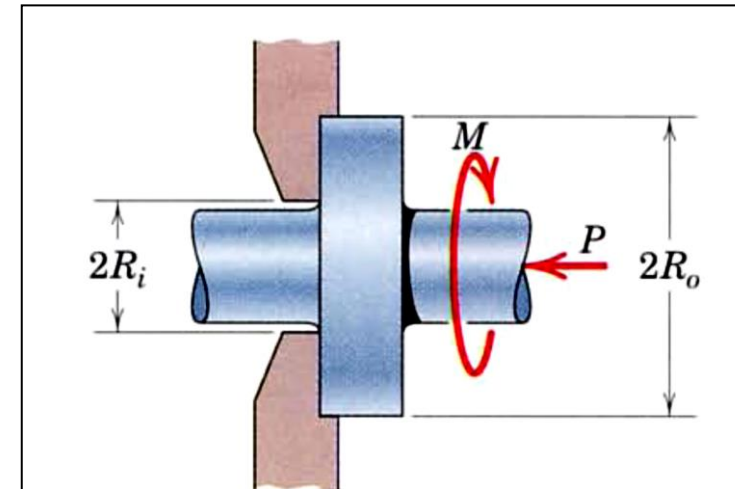
$$M = \frac{\mu P}{\pi R^2} \int_0^{2\pi} \int_0^R r^2 dr d\theta = \frac{2}{3} \mu P R$$



Moment in Collar (Uniform Pressure Theory)

- If the friction disks are rings, as in the collar bearing shown in Fig.

$$M = \frac{2}{3}\mu P \frac{R_o^3 - R_i^3}{R_o^2 - R_i^2}$$



Moment in Collar (Uniform Wear Theory)

❖ After the initial wearing-in period is over, the surfaces retain their new relative shape and further wear is therefore constant over the surface. This wear depends on both the circumferential distance traveled and the pressure p . Since the distance traveled is proportional to r , the expression $rp = K$ may be written, where K is a constant. The value of K is determined from the equilibrium condition for the axial forces, which gives

$$P = \int p \, dA = K \int_0^{2\pi} \int_0^R dr \, d\theta = 2\pi KR$$

With $pr = K = P/(2\pi R)$, we may write the expression for M as

$$M = \int \mu pr \, dA = \frac{\mu P}{2\pi R} \int_0^{2\pi} \int_0^R r \, dr \, d\theta$$

which becomes

$$M = \frac{1}{2}\mu PR$$

❖ **Note:** Wear is dependent on distance travel (which is dependent on radial distance ' r ') and applied pressure ' p '. 20

Moment in Collar (Uniform Wear Theory)

- ❖ The frictional moment for worn-in plates is, therefore, only $(\frac{1}{2})/(\frac{2}{3})$, or $\frac{3}{4}$ as much as for new surfaces. If the friction disks are rings of inside radius R_i and outside radius R_o , substitution of these limits gives for the frictional torque for worn-in surfaces

$$M = \frac{1}{2}\mu P(R_o + R_i)$$

- ❖ **Note:** Wear is dependent on distance travel (which is dependent on radial distance ‘ r ’) and applied pressure ‘ p ’.

Forces in Clutches

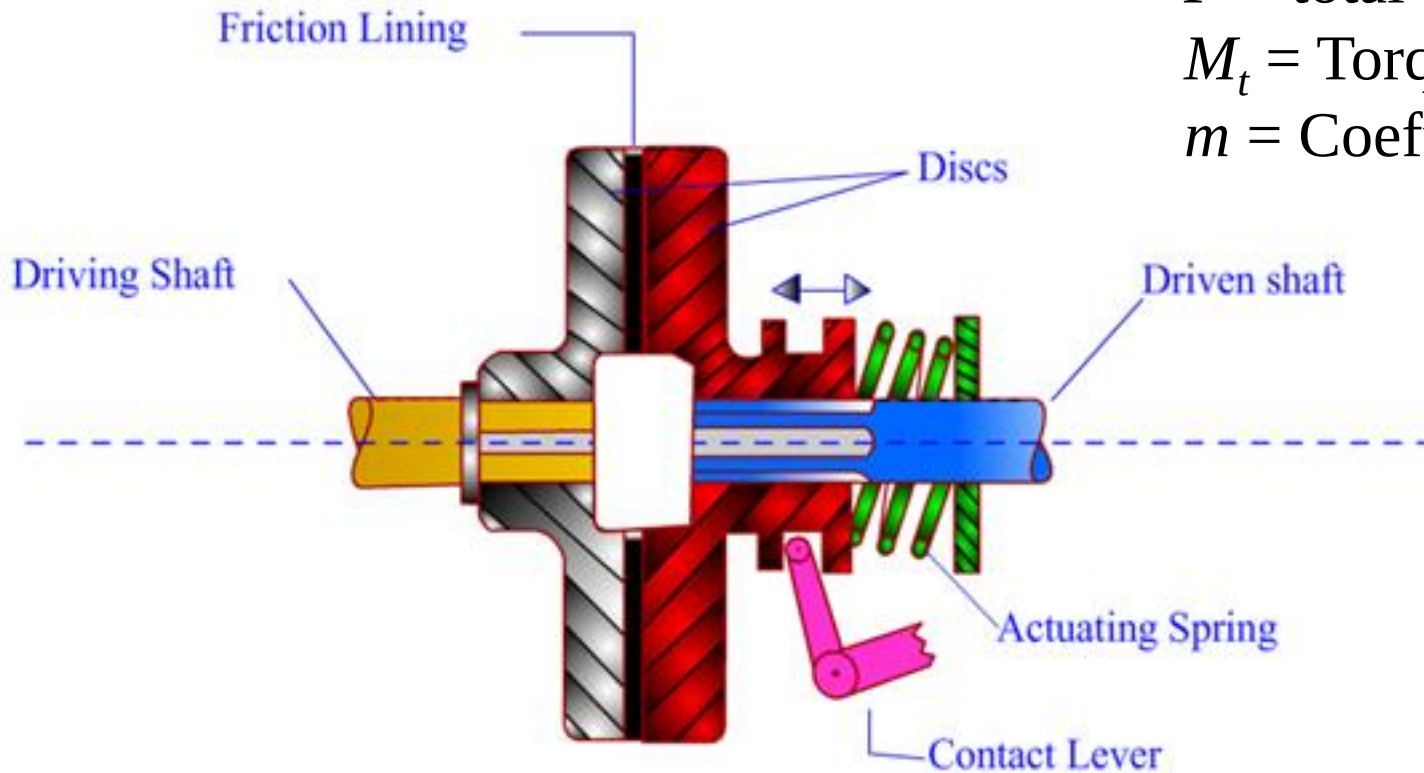


Fig1: Single friction disk clutch

F = total actuating force (axial force) (N)
 M_t = Torque transmitted by friction (N-mm)
 m = Coefficient of friction between friction disks

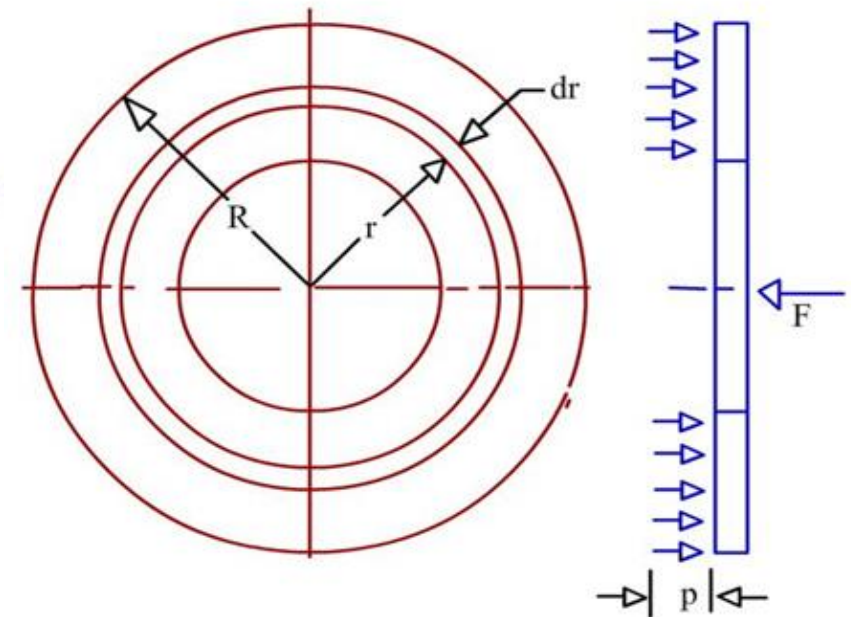


Fig2 :Notations of single disk friction clutch

Forces in Clutches

- ✓ Consider an elemental ring of radius, r and radial thickness dr .
- ✓ For this ring, the cross-sectional area of the element,

$$A = 2\pi r \cdot dr$$

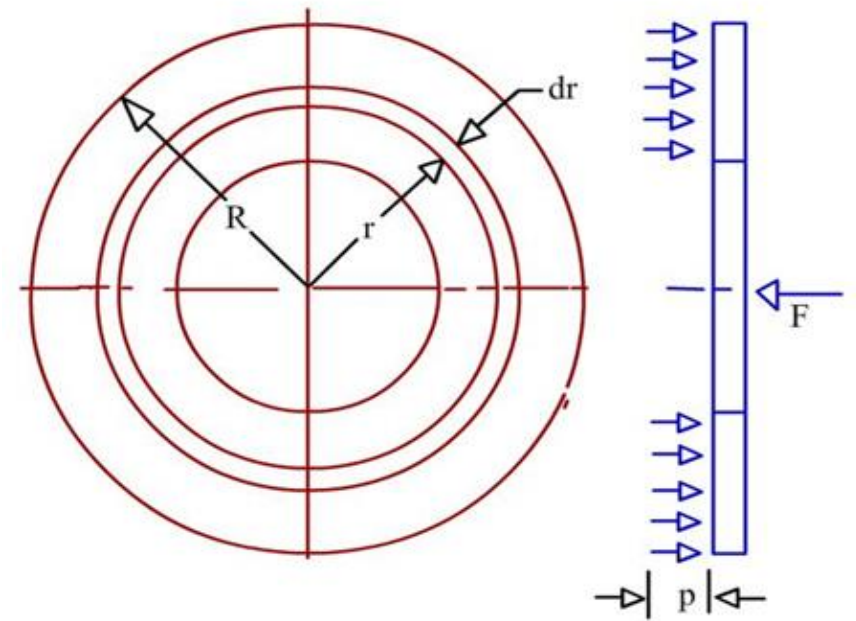
$$F = 2\pi r \cdot dr \cdot p$$

$$\text{Frictional force, } F_f = 2\pi r \cdot dr \cdot p \cdot \mu$$

$$M_t = 2\pi r^2 \cdot dr \cdot p \cdot \mu$$

$$F = 2\pi \int_r^R p r \cdot dr$$

$$M_t = 2\pi \cdot \mu \int_r^R p r^2 \cdot dr$$



1

2

Torque transmission under uniform pressure

- ❖ This theory is applicable to new clutches. In new clutches employing a number of springs, the pressure can be assumed as uniformly distributed over the entire surface area of the friction disk. With this assumption, the intensity of pressure between disks, p is regarded as constant. From Eq. (1) and (2)

$$F = \pi \cdot p (R^2 - r^2) \quad 3$$

$$M_t = \frac{2}{3} \pi \cdot \mu \cdot p (R^3 - r^3) \quad 4$$

Substituting Equation 3 in 4 we get,

$$M_t = \frac{2}{3} \mu \cdot F \frac{(R^3 - r^3)}{R^2 - r^2} \quad 5$$

- ✓ The above equation is valid for a single pair of mating disk surfaces

Torque transmission under uniform wear

This theory is based on the fact that wear is uniformly distributed over the entire surface area of friction disk. This assumption can be used for worn out clutches/old clutches. The axial wear of the friction disk is proportional to frictional work. The work done by the friction is proportional to the frictional force (μp) and the rubbing velocity ($2\pi r n$) where n is the speed of the disk in revolution per minute. When the speed n and the coefficient of friction m are constant for a given configuration, then

$$\text{Wear} \propto pr \quad 6$$

According to this assumption,

$$pr = \text{Constant} \quad 7$$

When the clutch plate is new and rigid, the wear at the outer radius will be more, which will reduce pressure at the outer edge due to rigid pressure plate. This will change pressure distribution.

Torque transmission under uniform wear

During running condition, the pressure distribution is adjusted such that the product ($p.r$) is constant. Therefore,

$$p.r = p_a.r \quad 8$$

Where p_a is the pressure at the inner edge of plate, which is also the maximum pressure. From equation (1) and (2)

$$F = 2\pi.p_a.r(R-r) \quad 9$$

From equation 2 and 8

$$M_t = \pi.\mu.p_a.r(R^2 - r^2) \quad 10$$

Substituting Π , P_a , r from equation 9 in the above equation

$$M_t = \mu.F\left(\frac{R+r}{2}\right) \quad 11$$

The above equation gives the torque transmitting capacity for a single pair of contacting surfaces.

Torque transmission under uniform wear

The uniform-pressure theory is applicable only when the friction lining is new. When the friction lining is used over a period of time, wear occurs. Therefore, the major portion of the life of friction lining comes under uniform-wear criterion. Hence, in the design of clutches, the uniform wear theory is used.

Problem on Disk Friction

Circular disk A (225 mm dia.) is placed on top of disk B (300 mm dia.) and is subjected to a compressive force of 400 N. Pressure under each disk is constant over its surface. Coefficient of friction between A and $B = 0.4$.

Determine:

- (a) the couple M which will cause A to slip on B .
- (b) Min coefficient of friction μ between B and supporting surface C which will prevent B from rotating shown in **Fig(e)**?

Solution:

✓ Impending slip between A and B :

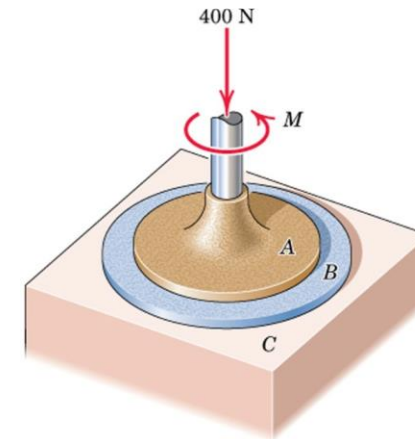
$$\mu = 0.4, P = 400 \text{ N}, R = 225/2 \text{ mm}$$

$$M = \frac{2}{3} \times 0.4 \times 400 \times 0.225/2 \quad \mathbf{M = 12 \text{ Nm}}$$

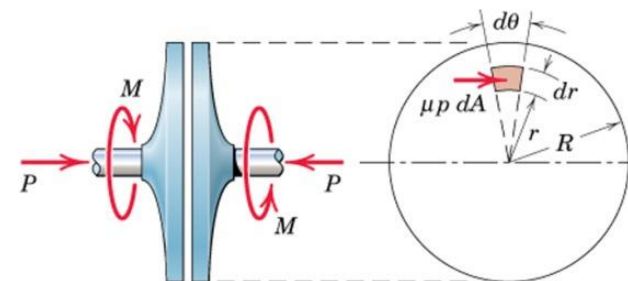
(b) Impending slip between B and C :
Slip between A and B $M = 12 \text{ Nm}$

$$\mu = ? \quad P = 400 \text{ N}, R = 300/2 \text{ mm}$$

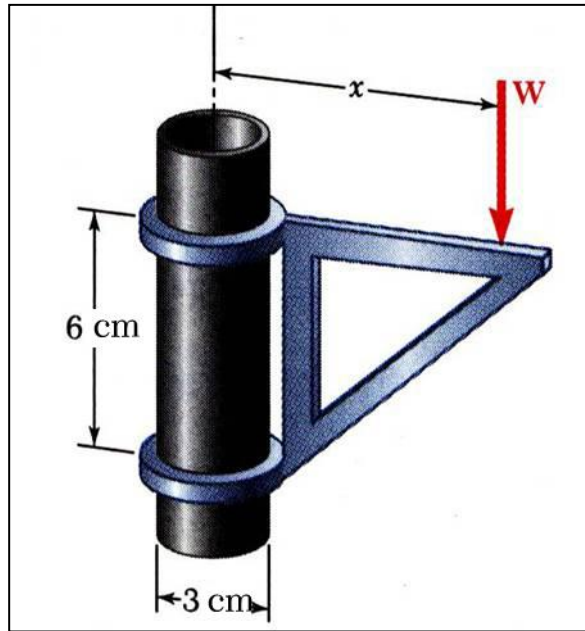
$$12 = \frac{2}{3} \times \mu \times 400 \times 0.300/2 \quad \mathbf{\mu = 0.3}$$



Fig(e)



Problem 8



Fig(d)

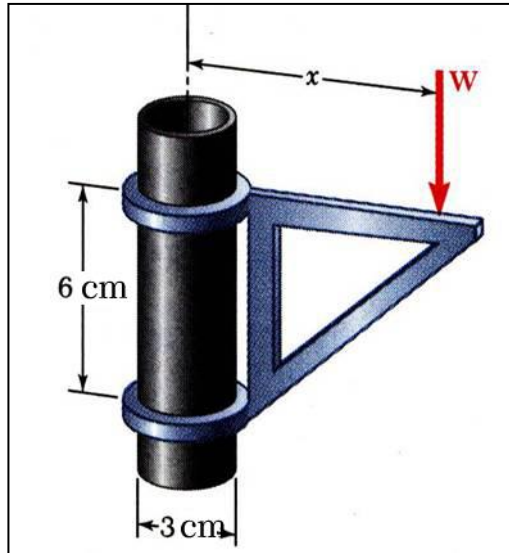
The moveable bracket shown may be placed at any height on the 3-cm diameter pipe. If the coefficient of friction between the pipe and bracket is 0.25, determine the minimum distance x at which the load can be supported. Neglect the weight^{8, 79} of the bracket in **Fig(d)**?

SOLUTION:

- ✓ When W is placed at minimum x , the bracket is about to slip and friction forces in upper and lower collars are at maximum value.
- ✓ Apply conditions for static equilibrium to find minimum x .

Problem 8 (contd.)

SOLUTION:

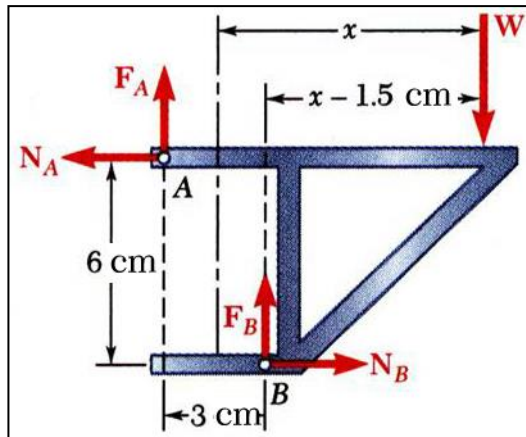


- ✓ When W is placed at minimum x , the bracket is about to slip and friction forces in upper and lower collars are at maximum value.

$$F_A = \mu_s N_A = 0.25 N_A$$

$$F_B = \mu_s N_B = 0.25 N_B$$

- ✓ Apply conditions for static equilibrium to find minimum x .

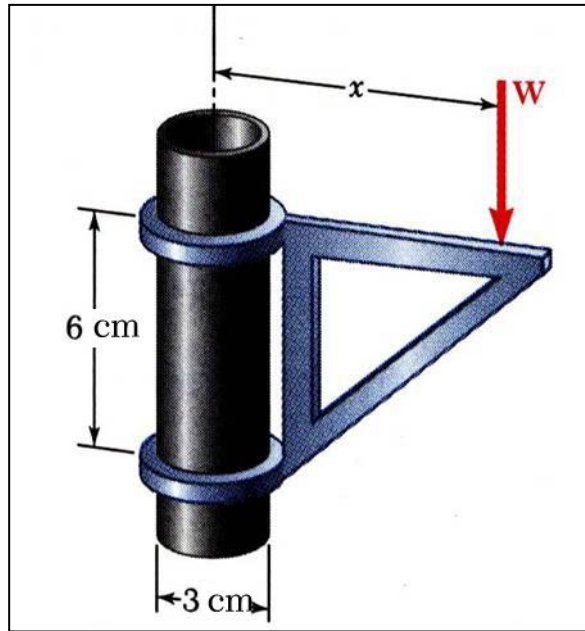


$$\begin{aligned}\sum F_x = 0: \quad N_B - N_A &= 0 & N_B &= N_A \\ \sum F_y = 0: \quad F_A + F_B - W &= 0 \\ 0.25 N_A + 0.25 N_B - W &= 0 \\ 0.5 N_A &= W & N_A = N_B &= 2W \\ \sum M_B = 0: \quad N_A(6 \text{ cm}) - F_A(3 \text{ cm}) - W(x - 1.5 \text{ cm}) &= 0 \\ 6 N_A - 3(0.25 N_A) - W(x - 1.5) &= 0 \\ 6(2W) - 0.75(2W) - W(x - 1.5) &= 0\end{aligned}$$

$$x = 12 \text{ cm}$$

End

Problem 8



Fig(d)

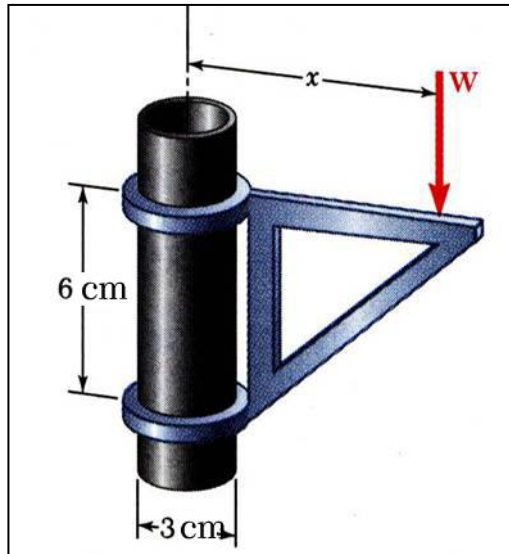
The moveable bracket shown may be placed at any height on the 3-cm diameter pipe. If the coefficient of friction between the pipe and bracket is 0.25, determine the minimum distance x at which the load can be supported. Neglect the weight⁸⁻³² of the bracket in **Fig(d)**?

SOLUTION:

- ✓ When W is placed at minimum x , the bracket is about to slip and friction forces in upper and lower collars are at maximum value.
- ✓ Apply conditions for static equilibrium to find minimum x .

Problem 8 (contd.)

SOLUTION:

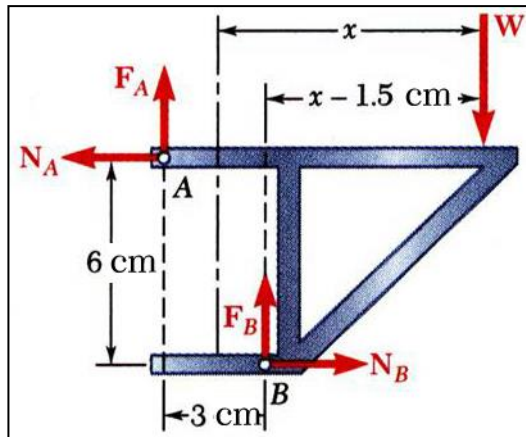


- ✓ When W is placed at minimum x, the bracket is about to slip and friction forces in upper and lower collars are at maximum value.

$$F_A = \mu_s N_A = 0.25 N_A$$

$$F_B = \mu_s N_B = 0.25 N_B$$

- ✓ Apply conditions for static equilibrium to find minimum x.



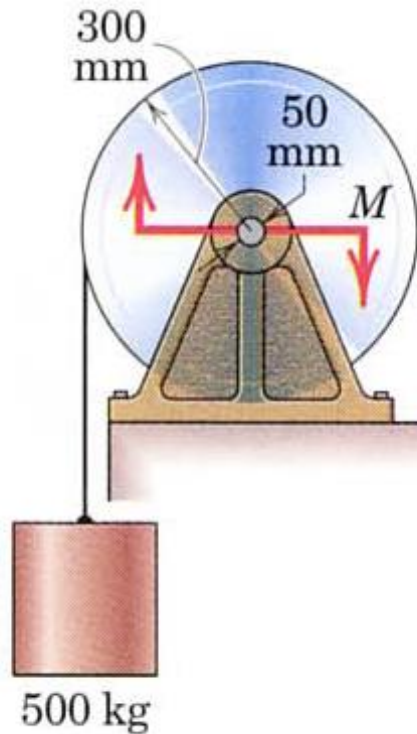
$$\begin{aligned}\sum F_x = 0: \quad N_B - N_A &= 0 & N_B &= N_A \\ \sum F_y = 0: \quad F_A + F_B - W &= 0 \\ 0.25 N_A + 0.25 N_B - W &= 0 \\ 0.5 N_A &= W & N_A = N_B &= 2W \\ \sum M_B = 0: \quad N_A(6 \text{ cm}) - F_A(3 \text{ cm}) - W(x - 1.5 \text{ cm}) &= 0 \\ 6 N_A - 3(0.25 N_A) - W(x - 1.5) &= 0 \\ 6(2W) - 0.75(2W) - W(x - 1.5) &= 0\end{aligned}$$

$$x = 12 \text{ cm}$$

Assignment

- 6/69** A torque M of $1510 \text{ N} \cdot \text{m}$ must be applied to the 50-mm-diameter shaft of the hoisting drum to raise the 500-kg load at constant speed. The drum and shaft together have a mass of 100 kg. Calculate the coefficient of friction μ for the bearing.

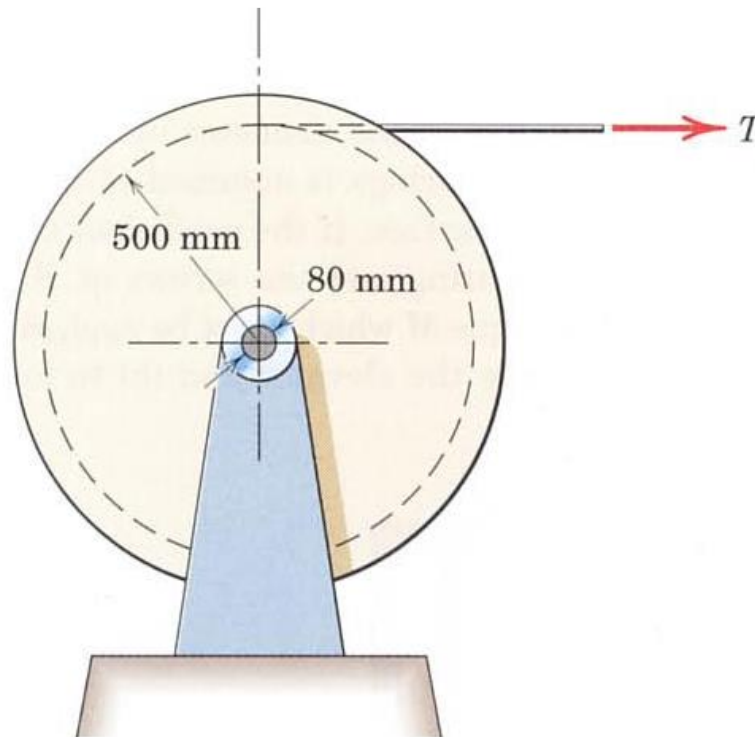
Ans. $\mu = 0.271$



Assignment

The telephone-cable reel has a mass of 250 kg and is mounted on an 80-mm-diameter shaft. If the coefficient of friction between the shaft and its bearing is 0.30, calculate the horizontal tension T required to turn the reel.

$$\text{Ans. } T = 56.4 \text{ N}$$

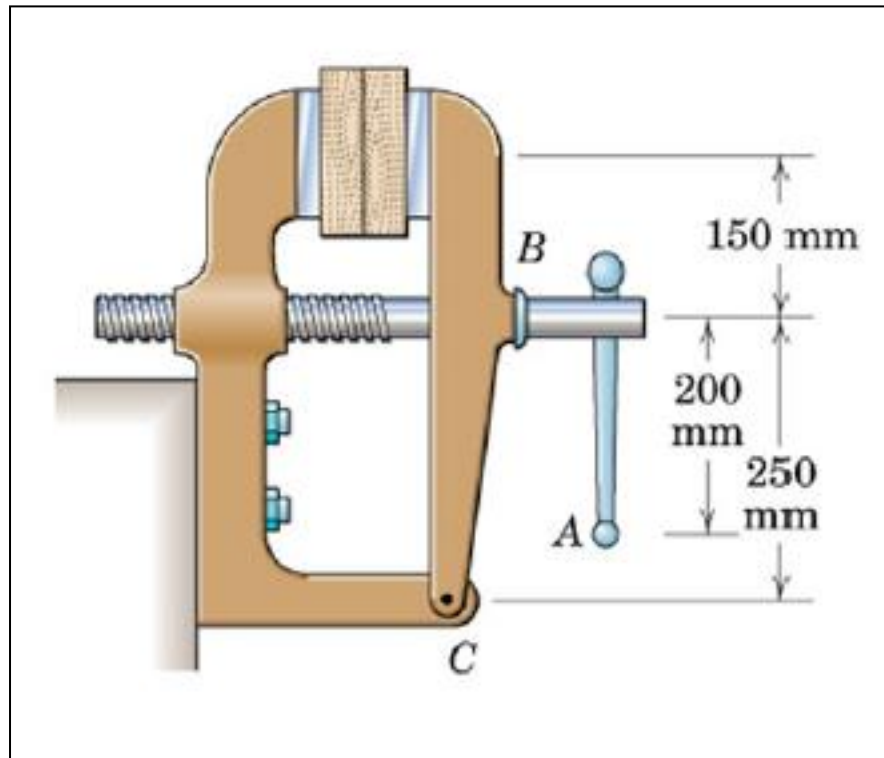


Problem 7

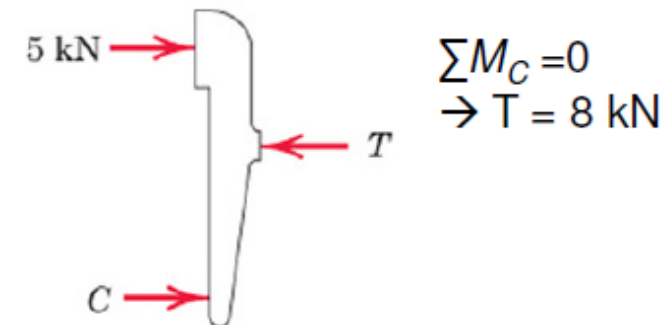
Single threaded screw of the vise has a mean diameter of 25 mm and a lead of 5 mm. A 300 N pull applied normal to the handle at A produces a clamping force of 5 kN between the jaws of the vise. Determine:

- (a) Frictional moment M_B developed at B due to thrust of the screw against body of the jaw
- (b) Force Q applied normal to the handle at A required to loosen the vise

μ_s in the threads = 0.20



Solution: Draw FBD of the jaw to find tension in the screw



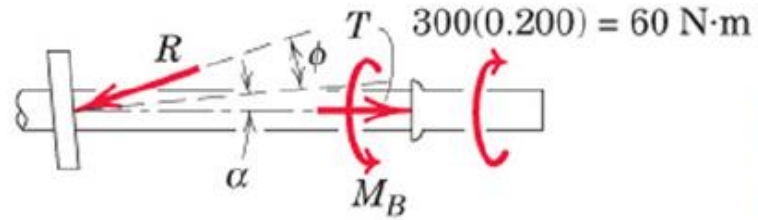
Find the helix angle α and the friction angle ϕ

$$\alpha = \tan^{-1} (L/2\pi r) = 3.64^\circ$$

$$\tan \phi = \mu \rightarrow \phi = 11.31^\circ$$

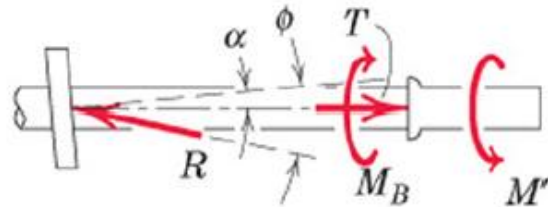
Problem 7 (contd.)

- (a) To tighten the vise
Draw FBD of the screw



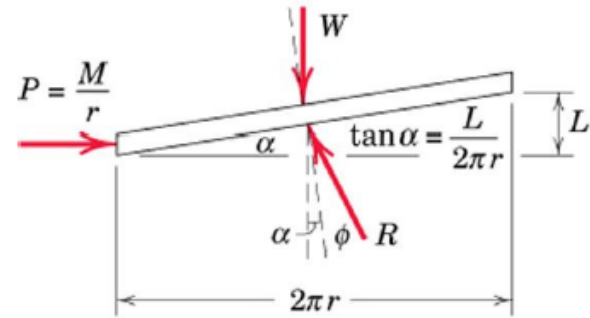
$$M = T r \tan(\alpha + \phi)$$
$$60 - M_B = 8000(0.0125)\tan(3.64 + 11.31)$$
$$\mathbf{M_B = 33.3 \text{ Nm}}$$

- (a) To loosen the vise (on the verge of being loosened)
Draw FBD of the screw: Net moment = applied moment M' minus M_B



$$M = T r \tan(\phi - \alpha)$$
$$M' - 33.3 = 8000(0.0125)\tan(11.31 - 3.64)$$
$$M' = 46.8 \text{ Nm}$$
$$\mathbf{Q = M'/d = 46.8/0.2 = 234 \text{ N}}$$

Moments for Unwinding Screw



To Raise Load

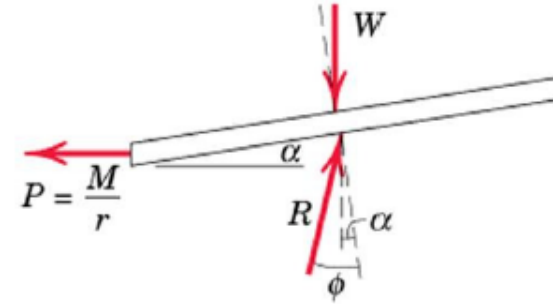
Equivalent force required to push the movable thread up the fixed incline is:

$$P = M/r$$

From Equilibrium:

$$M = W r \tan(\alpha + \phi)$$

If M is removed: the screw will remain in place and be self-locking provided $\alpha < \phi$ and will be on the verge of unwinding if $\alpha = \phi$



To Lower Load ($\alpha < \phi$)

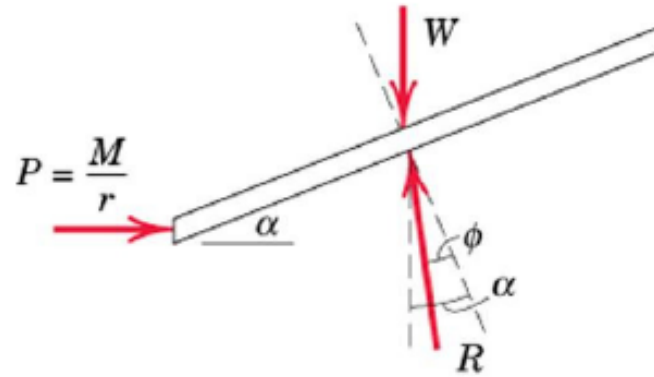
To lower the load by unwinding the screw, We must reverse the direction of M as long as $\alpha < \phi$

From Equilibrium:

$$M = W r \tan(\phi - \alpha)$$

→ This is the moment required to unwind the screw

Moments for Unwinding Screw by itself



To Lower Load ($\alpha > \phi$)

If $\alpha > \phi$, the screw will unwind by itself. Moment required to prevent unwinding:

From Equilibrium:

$$M = W r \tan(\alpha - \phi)$$

Demonstration of Raising of Cup by Screw Jack

